Validated Synthetic Patient Generation for Small Longitudinal Cohorts: Coagulation Dynamics Across Pregnancy

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Abstract

Small longitudinal clinical cohorts—common in maternal health, rare diseases, and early-phase trials—are often too small for robust computational modeling or machine learning. Generating synthetic patients that faithfully capture the joint, longitudinal, and mechanistic structure of real data would enable analyses that are otherwise infeasible. Here, we present a synthetic patient generation pipeline based on multiplicity-weighted Stochastic Attention (SA), a sampling framework rooted in modern Hopfield network theory that extends the standard Hopfield energy with per-pattern multiplicity weights, enabling continuous interpolation between unconditioned generation and targeted subpopulation amplification at inference time without retraining. We applied this framework to a longitudinal coagulation dataset collected from K=23 pregnant patients across three clinical visits (pre-pregnancy baseline, first trimester, and third trimester). By concatenating multi-visit patient profiles into unified patterns and sampling via Langevin dynamics on the multiplicity-weighted energy landscape, SA generated coherent longitudinal trajectories that preserved the geometry of the clinical data manifold. We validated synthetic patients through a four-level framework of increasing stringency: (1) marginal plausibility of individual features, (2) preservation of cross-visit covariance structure, (3) amplification of rare clinical subgroups (e.g., polycystic ovary syndrome, n=3 $\rightarrow$ 100) with preserved condition-specific signatures, and (4) mechanistic consistency with an independent 58-species ordinary differential equation model of the coagulation cascade. At all four levels, SA-generated patients were indistinguishable from real patients (83–92% cloud overlap under mechanistic validation), while a multivariate normal baseline failed at levels 2–4 due to its rank-deficient covariance estimate (n<p). These results demonstrate that geometry-preserving generative methods can produce clinically validated synthetic cohorts from very small longitudinal datasets.

Keywords: synthetic data generation, stochastic attention, modern Hopfield networks, multiplicity weighting, coagulation, pregnancy, longitudinal data, mechanistic validation

1 Introduction

In 2021, the United States maternal mortality rate reached 32.9 deaths per 100,000 live births—the highest level since 1965—with Black mothers dying at nearly three times the rate of White mothers [Toy, 2023, Harris, 2023]. Thrombotic embolisms and hemorrhage together accounted for more than 22% of these deaths [Trost et al., 2022], underscoring the central role of the coagulation system in maternal outcomes. Blood coagulation is a complex enzymatic cascade in which a small tissue factor (TF) stimulus triggers a series of serine protease activation reactions that ultimately produce thrombin, the central effector of hemostasis [Mann et al., 2011, Butenas et al., 2009]. Thrombin cleaves fibrinogen into fibrin, activates platelets, and amplifies its own production through positive feedback via Factors V, VIII, and XI, while simultaneously initiating its own shutdown through the thrombomodulin–protein C anticoagulant pathway [Hockin et al., 2002]. Pregnancy profoundly alters this hemostatic balance, shifting toward a hypercoagulable state that prepares for delivery [Hellgren, 2003, Grouzi et al., 2022, Bremme, 2003]: procoagulant factors including fibrinogen, Factor VIII, and von Willebrand factor increase substantially across gestation, while natural anticoagulants such as antithrombin and protein S decline. Conditions such as polycystic ovary syndrome (PCOS)—a hormonal and metabolic disorder affecting 6–12% of reproductive-age women [Azziz et al., 2004] that has been associated with secondary hemostatic changes including elevated Factor VIII and impaired fibrinolysis—and preeclampsia (PE)—a hypertensive disorder of pregnancy characterized by endothelial dysfunction, platelet activation, and altered thrombin generation that complicates 2–8% of pregnancies worldwide [Abalos et al., 2014]—further perturb this balance in ways that remain incompletely characterized. Mathematical models of the coagulation cascade, from the original Hockin–Mann stoichiometric framework [Hockin et al., 2002] through extensions incorporating the thrombomodulin–protein C pathway [Bravo et al., 2012] to the Brummel-Ziedins 2012 (BZ2012) model with detailed APC-mediated Factor Va inactivation kinetics [Brummel-Ziedins et al., 2012], have provided mechanistic insight into patient-specific thrombin generation [Luan et al., 2007, 2010], but applying these models to pregnancy cohorts requires longitudinal datasets capturing the full complement of coagulation factors, inhibitors, and thrombin generation assay (TGA) measurements across gestation—precisely the type of data that is rare and expensive to collect.

The fundamental barrier is sample size. Longitudinal pregnancy coagulation studies typically enroll tens of patients with complete follow-up across multiple trimesters, yielding datasets where the number of features p (coagulation factors, TGA parameters, viscoelastic measurements) exceeds the number of patients n. Rare complications compound this problem: small longitudinal cohorts inevitably contain only a handful of patients with any given condition, far too few for condition-specific statistical analysis. This $n < p$ regime fundamentally limits conventional approaches: covariance matrices are rank-deficient, cross-validation is unreliable, and overfitting is nearly inevitable. Synthetic data generation offers a path forward by augmenting small real datasets with statistically faithful synthetic records, enabling downstream analyses—mechanistic modeling, clinical comparisons, hypothesis generation for rare complications—that would otherwise be infeasible [Gonzales et al., 2023, Murtaza et al., 2024]. However, existing methods face fundamental limitations in the small-cohort longitudinal setting. Sampling from a fitted multivariate normal (MVN) distribution is singular when $n < p$ and must be regularized [Ledoit and Wolf, 2004], introducing bias that distorts the joint distribution; MVN also assumes Gaussian marginals, linear correlations, and provides no mechanism for conditional generation—it cannot amplify a specific subpopulation while preserving its signatures. More expressive models such as GANs and VAEs [Goodfellow et al., 2014, Kingma and Welling, 2014, Xu et al., 2019], including recent longitudinal extensions [Murtaza et al., 2024], require substantially larger training sets and are prone to mode collapse at small n

(we confirm this empirically for CTGAN at n=23 in this work).

Stochastic Attention (SA) provides an alternative framework rooted in modern Hopfield network theory [Ramsauer et al., 2021]—a continuous generalization of the associative memory model recognized by the 2024 Nobel Prize in Physics—in which stored patient profiles serve as memory patterns in a continuous energy landscape and Langevin dynamics generates novel samples that interpolate between them via the attention mechanism. We recently introduced a multiplicity-weighted extension of the Hopfield energy that adds per-pattern weights to the log-sum-exp term, enabling continuous interpolation between unconditioned generation and targeted subpopulation amplification at inference time without retraining [Varner, 2024]. Crucially, SA operates on the data manifold directly rather than fitting a parametric distribution, making it well-suited to the $n < p$ regime. In prior work, we generated synthetic patient populations using MVN distributions fitted independently to each clinical visit [Bravo et al., 2023]; while this captured per-visit statistics, it assumed Gaussian marginals and generated each visit independently rather than as part of a coherent longitudinal trajectory.

In this work, we adapted multiplicity-weighted SA for longitudinal clinical data and introduced a four-level validation framework to rigorously assess synthetic patient quality. We formulated a pipeline including concatenated-visit patterns, PCA dimensionality reduction, and a direction-magnitude decomposition that preserves anisotropic variance structure, with the multiplicity-weighted energy providing a unified mechanism for both unconditioned generation and targeted amplification of rare subgroups. We validated synthetic patients through four progressively stringent levels—marginal plausibility, cross-visit covariance preservation, conditional subgroup amplification, and mechanistic consistency with the BZ2012 ODE model [Brummel-Ziedins et al., 2012]—and demonstrated that SA-generated patients passed all four levels when applied to a 72-feature, 3-visit pregnancy coagulation dataset (K=23 patients, including 3 PCOS and 5 PE), while MVN failed at levels 2 through 4, and that the BZ2012 model confirmed 83–92% cloud overlap between real and synthetic patients under physiological conditions.

2 Methods

2.1 Clinical Dataset

We analyzed a longitudinal dataset of coagulation, fibrinolysis, thrombin generation assay (TGA), and rotational thromboelastometry (ROTEM) measurements collected from pregnant patients at three clinical visits: Visit 1 (pre-pregnancy baseline), Visit 2 (first trimester), and Visit 3 (third trimester). The dataset contained 153 total records across approximately 50 subjects. After removing columns with $> 30\%$ missing values and retaining only subjects with complete data across all three visits, we obtained $K = 23$ complete longitudinal profiles across $n = 72$ assay measurements per visit.

The 72 assay measurements spanned five categories: (i) hormones (estradiol, progesterone), (ii) coagulation factors and inhibitors (Factors II, V, VII, VIII, IX, X, XI, XII; AT, PC, TFPI, vWF, fibrinogen, etc.), (iii) fibrinolytic markers (plasminogen, PAI-1, PAI-2, α_2-AP, TAFI, tPA, D-dimer equivalents), (iv) thrombin generation parameters under four initiator conditions (TF at 5 pM, TF+TM, No TF, No TF+anti-FXIa), and (v) ROTEM/viscoelastic parameters (CT, MCF, alpha angle, LOT, ML, LT, AUC under TF Only and TF+tPA conditions). The dataset included three enrollment conditions: Healthy (n=14), prior preeclampsia (Prior PE, n=6), and polycystic ovary syndrome (PCOS, n=3). For conditional generation, we defined subgroups by clinical outcome rather than enrollment condition: Healthy (n=14), PCOS (n=3), and patients who developed PE during the study (n=5, partially overlapping with the Prior PE and PCOS enrollment groups).

2.2 Multiplicity-Weighted Stochastic Attention

We generated synthetic patients using a multiplicity-weighted variant of Stochastic Attention (SA) rooted in modern Hopfield network theory [Ramsauer et al., 2021]. We stored K memory patterns as columns of a matrix $\hat{\mathbf{X}} \in \mathbb{R}^{d \times K}$ and defined a weighted Hopfield energy landscape:

$$E_r(\boldsymbol{\xi}) = \frac{1}{2}\|\boldsymbol{\xi}\|^2 - \frac{1}{\beta}\log\sum_{k=1}^{K} r_k \exp\left(\beta\, \mathbf{m}_k^\top \boldsymbol{\xi}\right) \tag{1}$$

where $\{\mathbf{m}_k\}_{k=1}^K$ are stored memory patterns, β is an inverse temperature parameter controlling the sharpness of retrieval, and $\{r_k\}_{k=1}^K$ are per-pattern multiplicity weights. When all $r_k = 1$, this reduces to the standard modern Hopfield energy; when $r_k = \rho > 1$ for a designated subset of patterns, the energy landscape is continuously deformed to favor that subset. We sampled from this landscape using an Unadjusted Langevin Algorithm (ULA):

$$\boldsymbol{\xi}_{t+1} = (1-\alpha)\,\boldsymbol{\xi}_t + \alpha\,\hat{\mathbf{X}}\,\mathrm{softmax}\Big(\beta\,\hat{\mathbf{X}}^\top \boldsymbol{\xi}_t + \log \mathbf{r}\Big) + \sqrt{\frac{2\alpha}{\beta}}\,\boldsymbol{\varepsilon}_t \tag{2}$$

where α is a step size, $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $\log \mathbf{r}$ is the element-wise log of the multiplicity vector. The deterministic component drove $\boldsymbol{\xi}_t$ toward a multiplicity-weighted combination of stored patterns, while the stochastic component enabled exploration and novelty. The $\log r_k$ bias in the softmax logits provided a minimal, theoretically grounded mechanism for continuously interpolating between unconditioned generation ($\rho = 1$, all patterns equally weighted) and hard subset curation ($\rho \to \infty$, only designated patterns contribute). This is an inference-time operation that requires no retraining of the energy landscape.

The inverse temperature β controlled the trade-off between retrieval (large β, converging to nearest memory) and generation (small β, uniform mixing). We identified the critical β^* at the phase transition by computing the normalized attention entropy and finding the inflection point (maximum negative second derivative of $\bar{H}(\beta)$ in log-β space) over a logarithmic sweep of 80 values from $\beta = 0.1$ to 1000. This procedure is domain-agnostic: the phase transition is a geometric property of K patterns in d dimensions, not a property of the features they represent. For our dataset (K=23, d_{PCA}=18), the empirical inflection yielded $\beta^* \approx 2.94$, close to the theoretical prediction $\beta^* = \sqrt{d} \approx 4.24$. A sensitivity analysis across $\beta/\beta^* \in [0.1, 3.0]$ confirmed that the operating point balanced generation novelty against fidelity (Supplementary Table S1). For weighted sampling, the entropy calculation incorporated the multiplicity bias, yielding a ρ-dependent $\beta^*(\rho)$.

The effective number of patterns contributing to the dynamics was quantified by the participation ratio $K_{\mathrm{eff}} = (\sum_k r_k)^2 / \sum_k r_k^2$, which interpolated smoothly between the total pattern count (unconditioned) and the designated subset size (strongly conditioned). For a target effective fraction f_{target} of probability mass on the designated subset, the required multiplicity was $\rho = f_{\mathrm{target}} \cdot K_{\mathrm{bg}} / (K_{\mathrm{des}} \cdot (1 - f_{\mathrm{target}}))$, where K_{des} and K_{bg} are the designated and background pattern counts, respectively.

2.3 Pipeline for Continuous Longitudinal Data

Applying SA to continuous longitudinal clinical data required several adaptations beyond the standard Hopfield sampling framework. We first concatenated each patient's n=72 assay measurements across all three visits into a single vector of dimension $d_{\text{concat}} = 3n = 216$, so that each stored memory pattern encoded a complete longitudinal profile and generated synthetic patients would have internally consistent multi-visit trajectories. We then standardized each feature to zero mean and unit variance and applied PCA retaining 95% of the variance, reducing dimensionality from 216 to d_{PCA}=18. This compression served two purposes: it removed collinearity among the 216 features and yielded a memory-to-dimension ratio of $K/d_{\text{PCA}} \approx 1.28$, placing SA in a favorable operating regime where the number of stored patterns exceeded the dimensionality of the representation.

Standard SA operates on unit-norm patterns, which is appropriate for discrete data (e.g., one-hot protein sequences) but destroys the anisotropic variance structure of continuous clinical measurements—collapsing dispersion across all principal components to a unit sphere. We addressed this with a direction-magnitude decomposition: before normalization, we recorded each pattern's PCA-space norm $r_k = \|\mathbf{m}_k\|$; we then ran SA on unit-norm patterns to obtain a directional sample $\hat{\boldsymbol{\xi}}$, drew a magnitude r from the empirical distribution of $\{r_k\}$, and reconstructed the rescaled sample as $\boldsymbol{\xi}_{\text{rescaled}} = r \cdot \hat{\boldsymbol{\xi}}$. This preserved the directional structure learned by the Hopfield energy while restoring the natural scale of variation. The rescaled PCA vector was mapped back to the original feature space via inverse PCA and destandardization, then split into three 72-dimensional visit records.

To generate condition-specific cohorts (e.g., PCOS-only), we set the multiplicity $r_k = \rho$ for patterns belonging to the target subgroup and $r_k = 1$ for all others, and sampled using the weighted Langevin update (Eq. 2). The multiplicity ρ was computed to achieve a target effective fraction $f_{\text{target}} = 0.80$ of softmax attention on the designated subset, via $\rho = f_{\text{target}} \cdot K_{\text{bg}}/(K_{\text{des}} \cdot (1 - f_{\text{target}}))$. For the PCOS subgroup (K_{des}=3), this required $\rho \approx 26.7$, reducing the effective pattern count to $K_{\text{eff}} \approx 4.6$ (Supplementary Table S2). Magnitudes for the direction-magnitude decomposition were drawn from the condition-specific subset of norms, ensuring that the scale of variation matched the target subpopulation rather than the full cohort.

The complete generation pipeline is summarized in Algorithm 1. All experiments used step size $\alpha = 0.01$, $T = 2{,}000$ Langevin iterations per sample, and random seed 42 for reproducibility. At this step size, the Unadjusted Langevin Algorithm is valid without a Metropolis–Hastings acceptance correction. The pipeline was implemented in Julia (v1.12) using the packages listed in the code repository.

2.4 Baseline and Validation Framework

As a baseline, we generated synthetic patients by fitting a single multivariate normal (MVN) distribution to the same K=23 concatenated 216-dimensional patient profiles. Critically, both SA and MVN operated on the same concatenated representation—each patient's three visits were joined into one vector before generation, so that both methods had the opportunity to capture cross-visit dependencies. For SA, this structure was preserved through the Hopfield energy landscape operating on 18-dimensional PCA projections of the concatenated vectors. For MVN, the 216×216 sample covariance matrix had rank at most 22 and was regularized using Ledoit–Wolf shrinkage [Ledoit and Wolf, 2004], which inflated eigenvalues in the 194 null dimensions. We then evaluated both SA and MVN synthetic patients through four progressively stringent validation levels. **Level 1 (Marginal Plausibility)** tested whether individual feature distributions matched, using per-feature means, standard deviations, and known physiological relationships. **Level 2 (Joint Structure)** tested whether the cross-visit covariance was preserved, by comparing full 216 × 216 correlation matrices, eigenvalue spectra, and PCA projections between real, SA, and MVN populations. **Level 3 (Conditional Structure)** tested whether rare subgroups could be amplified while preserving condition-specific signatures, by generating cohorts of 100 patients each from the Healthy (n=14),

Algorithm 1 Multiplicity-Weighted SA for Longitudinal Patient Generation

Require: Patient data $\{\mathbf{x}_k^{(v)}\}$ for $k = 1, \ldots, K$ patients, $v = 1, 2, 3$ visits
Require: Multiplicity vector $\mathbf{r}$ (all ones for unconditioned; $r_k = \rho$ for designated subset)
Ensure: N synthetic longitudinal patient profiles

Concatenate: $\mathbf{x}_k \leftarrow [\mathbf{x}_k^{(1)}; \mathbf{x}_k^{(2)}; \mathbf{x}_k^{(3)}] \in \mathbb{R}^{216}$
Standardize: $\mathbf{z}_k \leftarrow (\mathbf{x}_k - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma}$
PCA: $\mathbf{m}_k \leftarrow \mathbf{W}^\top \mathbf{z}_k \in \mathbb{R}^{18}$ ▷ 95% variance retained
Record norms: $r_k^{\text{mag}} \leftarrow \|\mathbf{m}_k\|$; normalize $\hat{\mathbf{m}}_k \leftarrow \mathbf{m}_k / r_k^{\text{mag}}$
Find β^* via entropy inflection on $\hat{\mathbf{M}} = [\hat{\mathbf{m}}_1, \ldots, \hat{\mathbf{m}}_K]$
for $i = 1, \ldots, N$ **do**
  Initialize $\hat{\boldsymbol{\xi}}_0 \sim \text{Uniform}(\mathcal{S}^{17})$ ▷ random point on unit sphere
  for $t = 0, \ldots, T-1$ **do** ▷ Langevin dynamics
    $\hat{\boldsymbol{\xi}}_{t+1} \leftarrow (1-\alpha)\hat{\boldsymbol{\xi}}_t + \alpha \hat{\mathbf{M}}\, \text{softmax}(\beta^* \hat{\mathbf{M}}^\top \hat{\boldsymbol{\xi}}_t + \log \mathbf{r}) + \sqrt{2\alpha/\beta^*}\, \boldsymbol{\varepsilon}_t$
  end for
  Draw magnitude: $r \sim \{r_k^{\text{mag}}\}$ ▷ condition-specific if weighted
  Rescale: $\mathbf{m}_{\text{synth}} \leftarrow r \cdot \hat{\boldsymbol{\xi}}_T / \|\hat{\boldsymbol{\xi}}_T\|$
  Inverse PCA + destandardize: $\mathbf{x}_{\text{synth}} \leftarrow \boldsymbol{\sigma} \odot (\mathbf{W}\, \mathbf{m}_{\text{synth}} + \bar{\mathbf{z}}) + \boldsymbol{\mu}$
  Split into visit records: $\mathbf{x}_{\text{synth}}^{(1)}, \mathbf{x}_{\text{synth}}^{(2)}, \mathbf{x}_{\text{synth}}^{(3)}$
end for

PCOS (n=3), and PE (n=5) subgroups using multiplicity-weighted sampling. **Level 4 (Mechanistic Consistency)** tested whether synthetic patients produced biologically plausible outputs under an independent mechanistic model. We used the Hockin–Mann BZ2012 coagulation model [Hockin et al., 2002, Brummel-Ziedins et al., 2012], a system of 58 ODEs with 64 rate constants, in which 5 rate constants were calibrated on Visit 1 real patients at the population level and the remaining 59 were held at literature values. We ran the model on all real and synthetic patients under two conditions (TF-only and TF+TM) and compared predicted-versus-measured thrombin generation features, using cloud overlap—the fraction of synthetic predicted/measured ratios falling within the 5th–95th percentile range of real ratios—as the primary metric.

3 Results

We generated N=100 synthetic longitudinal patient profiles using SA—chosen to provide approximately 4× amplification over the real cohort while remaining within the regime where SA's attention mechanism produces diverse, non-degenerate samples—and compared them against the K=23 real patients across four progressively stringent validation levels. As a baseline, we also generated N=100 patients from a regularized multivariate normal (MVN) distribution fitted to the same data.

3.1 Marginal Plausibility

We first assessed whether SA-generated patients reproduced per-feature summary statistics across all three visits. The median mean relative error (MRE) across all 72 features was approximately 2%, with the majority of features below 5% (Table 1), indicating that SA faithfully captured the central tendencies of the real population. Representative coagulation factors—including Factor II (MRE < 1.5%), Factor VIII (MRE < 3%), antithrombin (MRE < 2%), and fibrinogen (MRE < 1.2%)—showed particularly strong agreement across all visits. MVN also performed well at this level (median MRE $\approx$ 2–3%), which was expected since moment-matching is guaranteed by construction. We verified that synthetic patients were not memorized copies of real patients using two complementary metrics. First, the mean novelty score (defined as $1-\max_k \cos(\hat{\boldsymbol{\xi}}, \hat{\mathbf{m}}_k)$) was 0.50 at $\beta = \beta^*$, indicating that generated samples were on average 50% away from their nearest stored pattern in angular distance (Supplementary Table S1). Second, we computed nearest-neighbor distances in standardized feature space: the median distance from each synthetic patient to their closest real patient was 6.14 (standardized units), compared with 6.70 for the median real-to-real nearest-neighbor distance—a ratio of 0.92, indicating that synthetic patients were approximately as far from real patients as real patients were from each other, with no evidence of memorization or privacy-compromising proximity.

We next examined whether known physiological relationships were preserved in the synthetic cohort. We plotted pairwise scatter plots of coagulation factor levels against thrombin generation parameters across all visits and observed that SA-generated patients occupied the same joint regions as real patients (Fig. 1). The expected inverse relationship between antithrombin and TF-initiated peak thrombin was preserved, as were the positive associations between Factor VIII and peak, prothrombin and ETP, and the inverse relationship between antithrombin and ETP. These results indicated that SA captured not only univariate distributions but also the biologically meaningful pairwise dependencies that underlie coagulation physiology.

We also evaluated whether the characteristic pregnancy-driven longitudinal trajectories were reproduced. We computed per-visit means and standard deviations for six key coagulation features and found that synthetic patients tracked the real trajectories closely, with overlapping confidence bands at all three visits (Fig. 2). Fibrinogen, Factor VIII, and von Willebrand factor showed the expected monotonic increases from baseline through the third trimester in both cohorts, while Factor II increased modestly and antithrombin declined then partially recovered, consistent with the known hypercoagulable shift of pregnancy. These patterns confirmed that the concatenated-visit SA pipeline preserved the directionality and magnitude of longitudinal change.

3.2 Cross-Visit Covariance Structure

We compared the full 216×216 cross-visit correlation matrices for real, SA-generated, and MVN-generated populations. The real data exhibited a characteristic block structure with strong within-visit correlations along the diagonal and structured off-diagonal blocks reflecting cross-visit dependencies—for example, a patient's Visit 1 Factor X level predicting their Visit 3 Factor X level. We found that SA preserved this block structure, including the off-diagonal cross-visit correlations (Fig. 3). The SA$-$Real residual matrix showed small, unstructured deviations distributed throughout, whereas the MVN$-$Real residual was notably smoother, reflecting the regularization-induced shrinkage of off-diagonal correlations toward zero. This difference was most apparent in the off-diagonal blocks, where MVN systematically underestimated cross-visit dependencies that SA retained.

We examined the eigenvalue spectrum of the sample covariance matrix to understand the structural origin of this difference. The real data had rank 22 (eigenvalues dropping to numerical zero at component 23), reflecting the K=23 patient constraint. SA's spectrum dropped at component 18 (the PCA truncation point), faithfully reflecting the low-rank structure without introducing variance in null dimensions. In contrast, MVN's Ledoit–Wolf regularization inflated eigenvalues beyond component 22, producing spurious variance in 194 dimensions where the data contained none (Supplementary Fig. S3). This spurious variance manifested as the increased dispersion visible in PCA projections: SA-generated patients occupied the same region of PCA space as real patients across all three visits, while MVN-generated patients showed substantially greater scatter, particularly at Visits 2 and 3 (Fig. 4).

3.3 Conditional Generation of Rare Subgroups

We tested whether SA could amplify rare clinical subpopulations while preserving their condition-specific signatures. Our dataset contained three clinical subgroups—Healthy (n=14), PCOS (n=3), and PE (n=5)—of which the PCOS and PE groups were too small for any independent statistical analysis. We used SA's multiplicity-weighted sampling to generate condition-specific cohorts of 100 patients each by upweighting attention on the relevant stored patterns.

We compared condition-specific feature means between real and SA-generated patients across eight coagulation features that exhibited between-condition variation (Fig. 5). SA-generated cohorts preserved the condition-specific patterns: PCOS patients showed elevated Factor VIII and vWF relative to Healthy patients in both real and synthetic data, PE patients showed elevated α_2-AP and Factor IX, and the between-condition rank ordering was maintained across all eight features. To quantify equivalence, we performed a bootstrap Mann–Whitney test: for each feature and condition, we subsampled the synthetic cohort to match the real sample size (n_{real}), applied a Mann–Whitney U test, and repeated 1,000 times. We reported the fraction of replicates where $p > 0.05$ (i.e., real and synthetic were statistically indistinguishable). Across all 24 feature–condition pairs, 19 (79%) achieved $p > 0.05$ in $\geq$90% of replicates, with a median non-significance fraction of 96.2%. The five features that fell below 90% were concentrated in high-variance features (FIX, FVII, fibrinogen) where the real data exhibited large inter-patient variability. No multiple-testing correction was applied to the 24 simultaneous comparisons; the failing features had consistently low non-significance fractions (68–88%) rather than borderline values, so correction would not change the classification. PCA projections of the conditioned cohorts are provided in Supplementary Figs. S5–S6. These results demonstrated that SA's multiplicity-weighted interpolation on the data manifold can meaningfully amplify rare subgroups in a way that preserves their distinguishing clinical features—a capability with direct relevance to hypothesis generation and power analysis in understudied patient populations. MVN has no mechanism for this: fitting a separate MVN to three PCOS patients across 216 features is mathematically impossible (rank 2 covariance), and post-hoc subsetting of unconditional MVN samples does not preserve subgroup-specific structure.

3.4 Mechanistic Consistency

As a final, orthogonal validation, we tested whether SA-generated patients produced biologically plausible outputs when their coagulation factor levels were fed through an independent mechanistic model that knows nothing about the SA generation process. We used the Hockin–Mann BZ2012 model [Hockin et al., 2002, Brummel-Ziedins et al., 2012], a system of 58 ordinary differential equations with 64 rate constants that simulates thrombin generation from patient-specific coagulation factor inputs. We calibrated 5 of the 64 rate constants—prothrombinase, intrinsic and extrinsic tenase, protein C activation, and meizothrombin conversion (Supplementary Table S4)—on Visit 1 real patients at the population level (not per-patient), held the remaining 59 at published literature values, and ran the calibrated model on all 23 real patients (3 visits $\times$ 2 conditions = 138 simulations, 0 failures) and all 100 synthetic patients (3 visits $\times$ 2 conditions = 600 simulations, 0 failures). For each patient, we computed five ODE-predicted thrombin generation features (lagtime, peak, time-to-peak, maximum rate, and ETP) and compared them against the corresponding values from the patient record. In these comparisons (Fig. 6, top row), the horizontal axis is the TGA value from the patient record—which exists for both real and synthetic patients—while the vertical axis is the value computed by the BZ2012 model from that patient's factor inputs. Systematic deviations from the y=x line reflect ODE model bias (e.g., lagtime is underpredicted at $\approx$0.77$\times$), not a failure of the synthetic data; the key observation is that real and synthetic patients share the same pattern of model bias, occupying the same cloud with the same systematic offset.

We formalized this observation by computing the ODE-predicted/dataset ratio for each patient and comparing the resulting distributions between real and synthetic populations (Fig. 6, bottom row). These ratio distributions capture how the ODE model processes each patient's factor levels: if SA had generated patients with biologically implausible factor combinations, the model would process them differently, producing a shifted or broadened ratio distribution. Instead, the distributions overlapped substantially, with cloud overlap (fraction of synthetic ratios within the 5th–95th percentile of real ratios) ranging from 83% for ETP to 92% for lagtime, and two-sample Kolmogorov–Smirnov tests confirmed that the ratio distributions were statistically indistinguishable across all five TGA features (KS $D = 0.081$–0.123, all $p > 0.35$). Under TF+TM conditions, the model systematically overcorrected the protein C anticoagulant pathway, producing peak and maximum rate predictions approximately $0.5\times$ measured values for both real and synthetic patients (Supplementary Fig. S7), yet this shared systematic bias was itself informative—cloud overlap remained high at 88–92%, confirming that the model processed real and synthetic patients identically even under conditions where the calibration was imperfect.

The calibration also generalized across pregnancy timepoints: although fitted only on Visit 1 real patients, the predicted-to-measured ratios at Visits 2 and 3 under TF-only conditions were $0.96\times$ and $1.07\times$, respectively (Supplementary Fig. S9), confirming that the calibrated rate constants captured a stable population-level property of the coagulation system rather than overfitting to the training visit. Spearman rank correlations between predicted and measured values were moderate for ETP ($\rho \approx 0.6$–0.8 for real, $\rho \approx 0.6$–0.65 for synthetic; Supplementary Fig. S10) and weak for other features, consistent with a 5-parameter population-level calibration that was not designed to resolve inter-patient variability. The key finding of Level 4 was not patient-level prediction but population-level plausibility: an independent mechanistic model encoding decades of coagulation biochemistry confirmed that SA-generated patients fell within the same biologically plausible envelope as real patients, and could not be distinguished from them by the ODE model under either experimental condition.

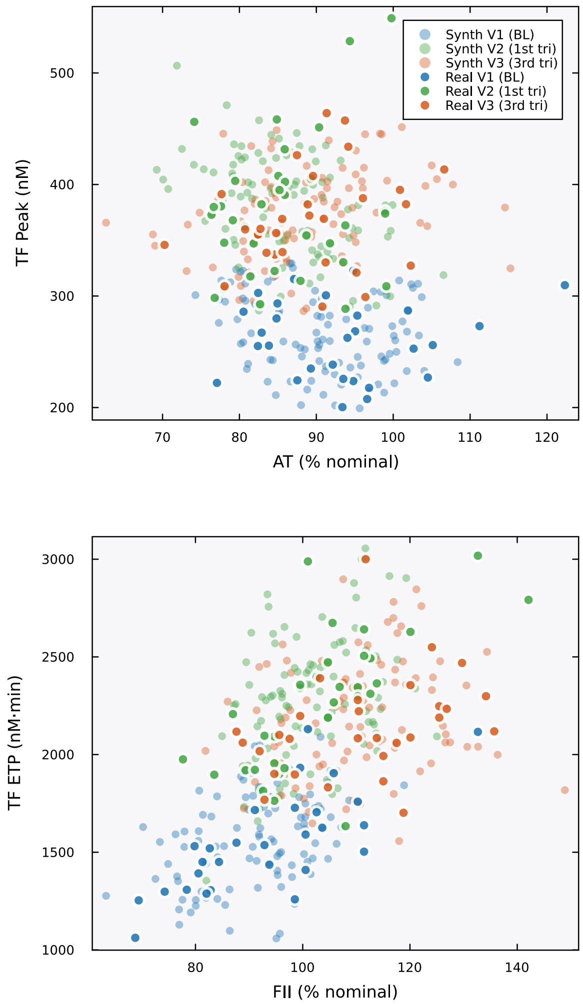

Figure 1: **Physiological correlations are preserved in synthetic patients.** Scatter plots of coagulation factor levels versus thrombin generation parameters across all three visits. Filled markers are real patients; open markers are SA-generated synthetic patients. Colors encode visit: blue (V1/baseline), green (V2/first trimester), orange (V3/third trimester). Real and synthetic patients occupy the same joint regions across all four pairwise relationships, confirming that biologically meaningful dependencies are preserved: AT inversely correlates with TF-initiated peak thrombin (top left), FVIII positively correlates with peak (top right), and both FII (bottom left) and AT (bottom right) show the expected relationships with ETP. The visit-dependent shift in these relationships—reflecting the pregnancy-driven increase in procoagulant factors—is also preserved in the synthetic cohort.

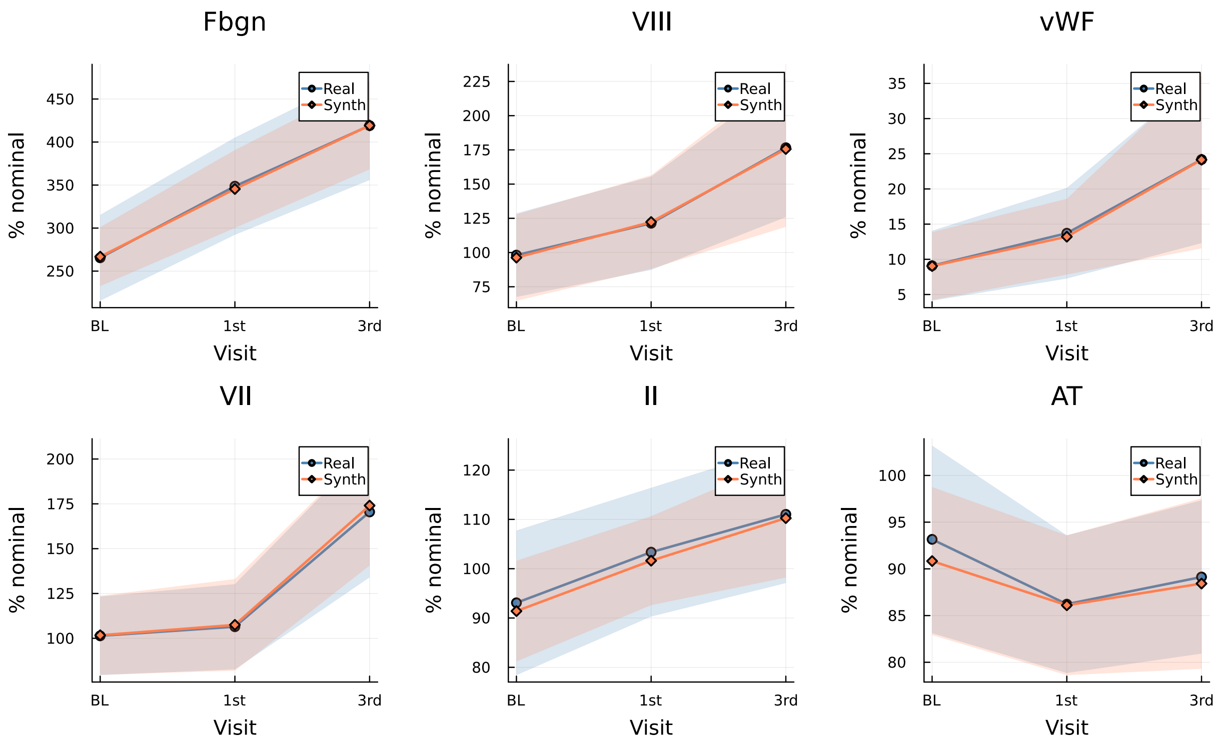

Figure 2: **Pregnancy-driven longitudinal trajectories are faithfully reproduced.** Mean ± standard deviation bands for six key coagulation features across visits (BL = baseline, 1st = first trimester, 3rd = third trimester) for real (blue) and SA-generated synthetic (orange) patients. The characteristic pregnancy-driven increases in fibrinogen, Factor VIII, and vWF are captured, as are the stable-to-declining patterns in Factor II and AT.

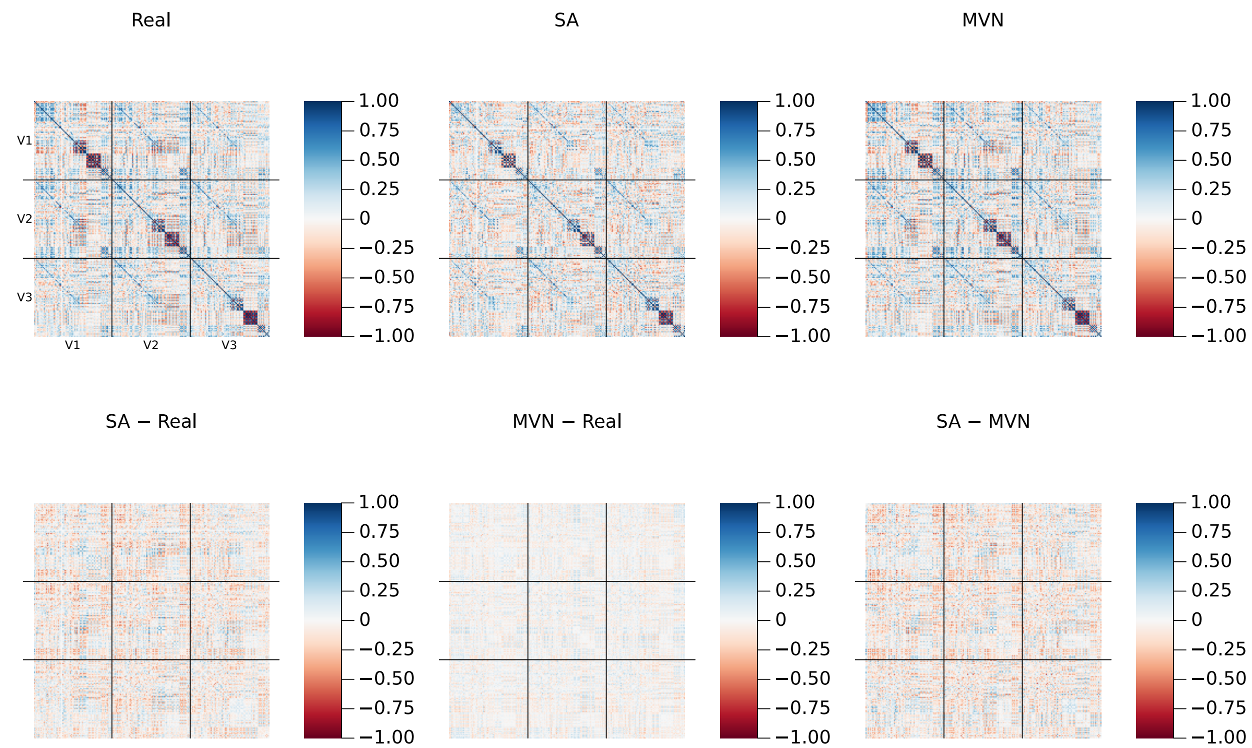

Figure 3: **Cross-visit correlation structure.** Top row: full 216×216 Pearson correlation matrices for Real (K=23, left), SA (N=100, center), and MVN (N=100, right) populations. Each matrix is organized as a 3×3 grid of 72×72 blocks (delineated by black lines), where the diagonal blocks capture within-visit feature correlations (V1–V1, V2–V2, V3–V3) and the off-diagonal blocks capture cross-visit dependencies (e.g., V1–V3: does a patient's baseline Factor VIII predict their third-trimester Factor VIII?). SA preserves both the within-visit and cross-visit block structure. Bottom row: pairwise residual matrices (element-wise subtraction), all plotted on the same ± 1 color scale as the top row. The SA−Real residual shows small, unstructured deviations distributed throughout. The MVN−Real residual is notably smoother—particularly in the off-diagonal blocks—reflecting the Ledoit–Wolf regularization that systematically shrinks cross-visit correlations toward zero. The SA−MVN residual confirms that the two methods differ most in precisely these off-diagonal blocks, where SA retains longitudinal dependencies that MVN's rank-deficient covariance estimate cannot represent. A version with amplified residual color scale (± 0.5) and Frobenius norms is provided in Supplementary Fig. S1.

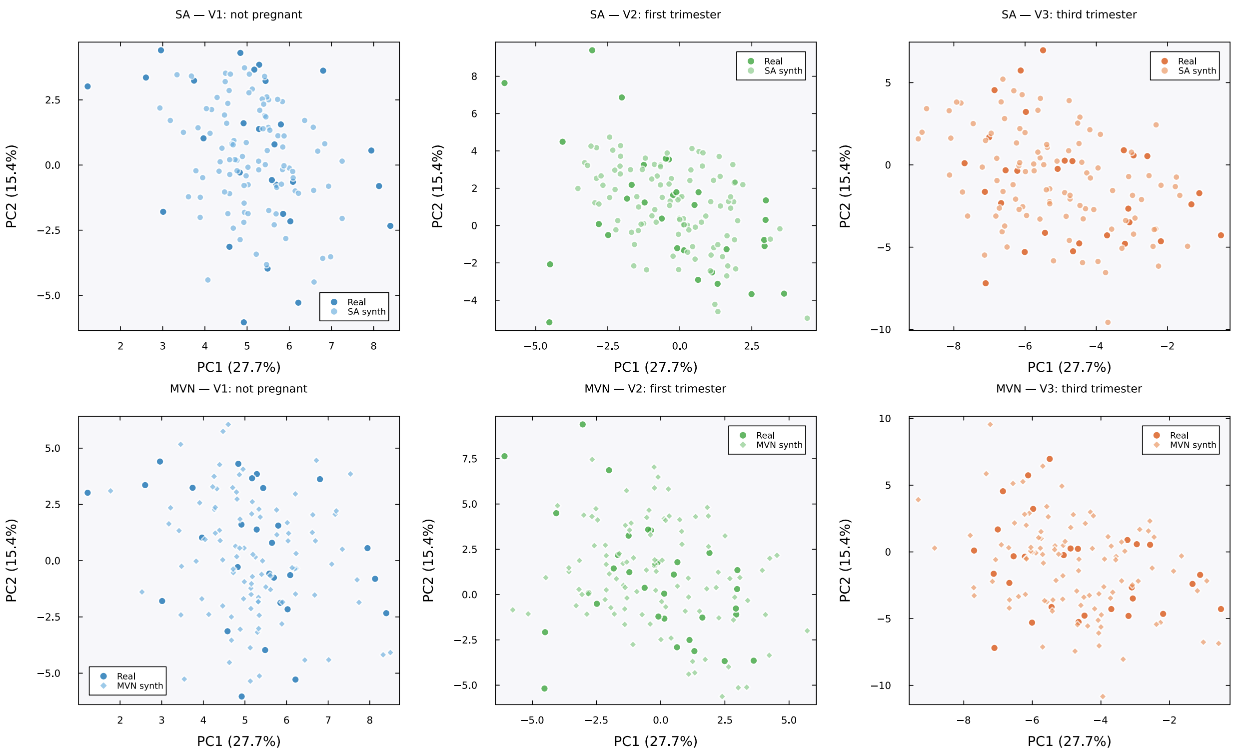

Figure 4: **PCA projections by visit: SA vs. MVN.** Each panel shows the first two principal components (PC1: 27.7% variance, PC2: 15.4%) computed from standardized per-visit real patient data (K=23 patients per visit, 72 features). Dark markers indicate real patients; lighter markers indicate synthetic patients (N=100). Top row: SA-generated patients (circles) cluster tightly around the real data cloud at all three visits, indicating that SA preserves the geometry of the data manifold. Bottom row: MVN-generated patients (diamonds) show substantially greater dispersion, particularly at Visits 2 (first trimester) and 3 (third trimester), where MVN patients extend well beyond the convex hull of the real data. This excess scatter reflects the spurious variance introduced by Ledoit–Wolf regularization of the rank-deficient covariance (Supplementary Fig. S3). Colors encode visit: teal (V1, not pregnant), green (V2, first trimester), orange (V3, third trimester).

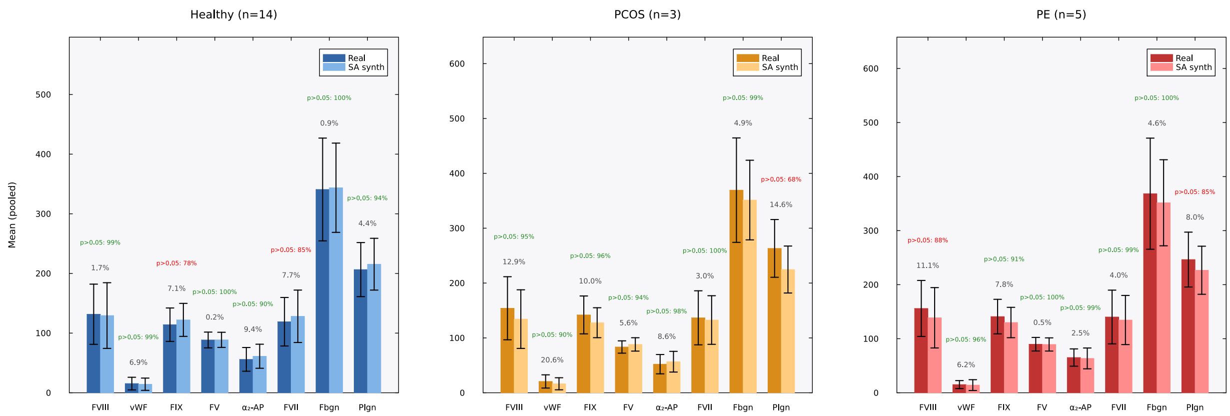

Figure 5: **Condition-specific feature preservation.** Grouped bar charts comparing real (dark bars) and SA-generated synthetic (light bars) patient means (± SD error bars) for eight coagulation features, shown separately for each clinical subgroup: Healthy (n=14, left, blue), PCOS (n=3, center, orange), and PE (n=5, right, red). Values are pooled across all three visits. Gray percentages indicate the mean relative error (MRE). Green annotations show the bootstrap Mann–Whitney equivalence result: the fraction of 1,000 bootstrap replicates (synthetic subsampled to n_{real}) in which $p > 0.05$, i.e., real and synthetic could not be distinguished. Features achieving ≥90% are shown in green; <90% in red. Across all 24 feature–condition pairs, 19 (79%) were statistically indistinguishable in ≥90% of replicates (median: 96.2%). The five features below this threshold were concentrated in high-variance features (FIX, FVII, fibrinogen) where large inter-patient variability reduced test power.

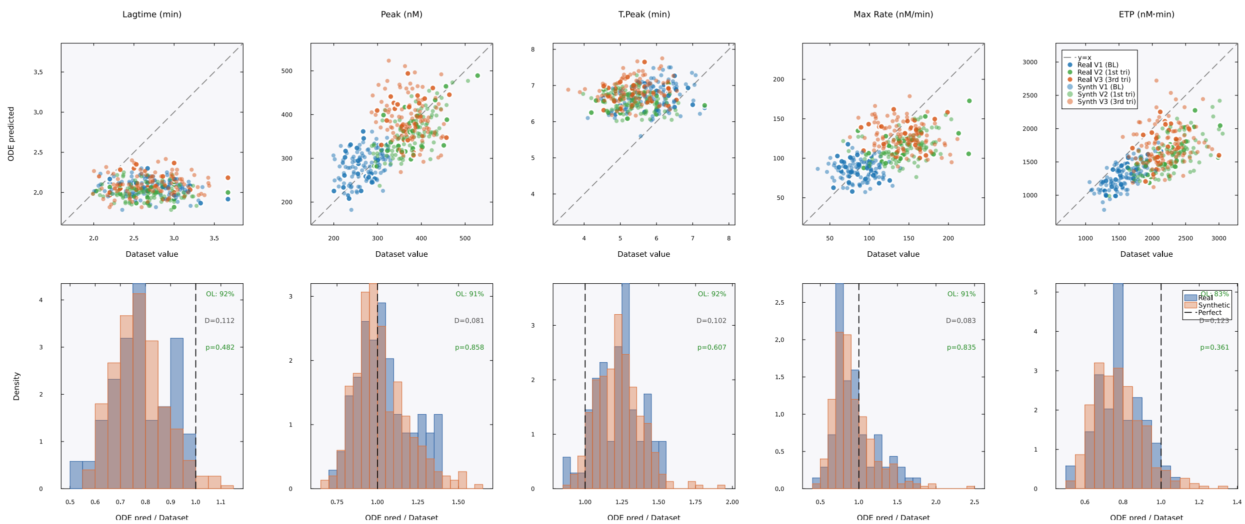

Figure 6: **Mechanistic validation (TF-only).** Top row: BZ2012 ODE-predicted TGA values (vertical axis) versus dataset TGA values (horizontal axis) for five thrombin generation parameters. The dataset values are the TGA measurements from each patient's record—present for both real and synthetic patients. The ODE-predicted values are computed by running each patient's coagulation factor levels through the 58-species BZ2012 model. The dashed line indicates where the ODE model would perfectly reproduce the dataset value ($y{=}x$); systematic deviations from this line (e.g., lagtime underprediction) reflect ODE model bias, not synthetic data failure. Filled markers are real patients, open markers are SA-generated synthetic patients, colored by visit: blue (V1/baseline, training set), green (V2/first trimester), orange (V3/third trimester). The key observation is that real and synthetic patients occupy the same cloud with the same pattern of ODE model bias. Bottom row: ODE-predicted/dataset ratio distributions for real (blue) and synthetic (orange) patients. If SA had generated biologically implausible factor combinations, the ODE model would process them differently, producing divergent ratio distributions. Instead, the distributions overlap substantially. Each panel is annotated with cloud overlap (OL), the two-sample Kolmogorov–Smirnov statistic (D), and p-value. All five features show $p > 0.35$, confirming that the ODE model cannot distinguish real from synthetic patients.

Table 1: Per-visit mean relative error (MRE) between real and SA-generated synthetic patients for representative coagulation features, computed as $|\bar{x}_{\text{synth}} - \bar{x}_{\text{real}}|/|\bar{x}_{\text{real}}|$.

Feature	Visit 1	Visit 2	Visit 3
Factor II (%)	0.009	0.012	0.015
Factor VIII (%)	0.017	0.030	0.025
AT (%)	0.013	0.018	0.014
Fibrinogen (mg/dL)	<0.001	0.008	0.012
TF Peak (nM)	0.022	0.019	0.027
TF ETP (nM·min)	0.015	0.021	0.018

4 Discussion

The central question of this work was whether a geometry-preserving generative method could produce synthetic patients from a very small longitudinal cohort that were not merely statistically similar to real patients but consistent with the underlying coagulation biology. Our results provided evidence at multiple levels of resolution that SA-generated patients met this standard. Individual feature distributions were faithfully reproduced, which MVN also achieved by construction; but SA additionally preserved the joint cross-visit covariance structure that MVN's rank-deficient estimate could not represent—a difference made visible in the eigenvalue spectrum, where MVN introduced spurious variance in 194 null dimensions while SA respected the low-rank geometry of the data. SA could amplify clinically meaningful rare subgroups—generating 100 synthetic PCOS patients from only 3 real ones—while preserving subgroup-specific signatures, a capability that MVN fundamentally lacks. And an independent ODE model encoding decades of coagulation biochemistry, calibrated exclusively on real patients, confirmed that the factor-to-thrombin-generation mapping in synthetic patients was biologically plausible, with 83–92% cloud overlap and Kolmogorov–Smirnov tests unable to distinguish the two populations ($p > 0.35$ for all five TGA features).

The ability to generate validated synthetic cohorts from very small longitudinal datasets has immediate practical implications for maternal health research. Rare pregnancy complications such as PCOS, preeclampsia, and gestational thrombophilia are difficult to study because assembling large, well-characterized longitudinal cohorts requires years of recruitment across multiple clinical sites. Our results suggested that SA can enable hypothesis generation, power analyses, and computational modeling studies for these understudied populations by generating synthetic cohorts that preserve both the statistical and mechanistic properties of the real data. The conditional generation capability is particularly relevant: rather than requiring researchers to collect additional rare patients, SA can amplify existing small subgroups into cohorts large enough for meaningful analysis.

The key insight underlying SA's advantage over parametric alternatives is geometric. SA generates samples by interpolating between stored patient profiles via the Hopfield energy landscape operating in a PCA-reduced linear subspace, rather than fitting a parametric distribution that must regularize or truncate high-dimensional structure. In the $n < p$ regime that characterizes most small clinical studies, any parametric distribution faces a fundamental identifiability problem: there are more parameters to estimate than observations to estimate them from. SA sidesteps this by operating in a PCA-reduced space where the memory-to-dimension ratio is favorable ($K/d_{\text{PCA}} \approx 1.28$) and by preserving the full directional structure of the data through the attention mechanism. A natural concern is whether the β^* selection procedure—originally developed for protein sequence generation [Varner, 2024]—transfers to clinical coagulation data. The entropy inflection method identifies the phase transition in the Hopfield energy landscape, which is a geometric property of K patterns in d dimensions, not a property of what the features represent; for our dataset the empirical inflection ($\beta^* \approx 2.94$) was close to the theoretical prediction ($\sqrt{d} \approx 4.24$), and a sensitivity analysis confirmed that generation quality degraded gracefully across $\beta/\beta^* \in [0.1, 3.0]$ (Supplementary Table S1). Similarly, varying the PCA variance retention threshold from 85% to 99% ($d_{\mathrm{PCA}} = 13$ to 21) had minimal impact on generation quality: median MRE remained between 1.2% and 1.8% across all thresholds (Supplementary Table S7), confirming that the 95% threshold was not a critical design choice. The multiplicity weighting for conditional generation introduced an additional consideration: achieving 80% attention on the PCOS subgroup required $\rho \approx 26.7$, reducing the effective pattern count to $K_{\mathrm{eff}} \approx 4.6$ (Supplementary Table S2), which explains the larger MREs observed for PCOS features—the energy landscape was effectively shaped by fewer than 5 independent patterns, pulling means toward the population average.

The mechanistic validation introduced a paradigm that we believe is broadly applicable beyond coagulation: rather than relying solely on statistical comparisons between real and synthetic distributions, we tested whether synthetic patients produced biologically consistent outputs when processed by an independent computational model. This approach is available whenever a validated mechanistic model exists for the domain of interest—pharmacokinetic models in drug development, physiological models in critical care, or metabolic models in oncology—and the requirement is not that the model be a perfect predictor, but that synthetic patients fall within the same model-predicted envelope as real patients. The bootstrap Mann–Whitney analysis of condition-specific features revealed that 5 of 24 (21%) feature–condition pairs were statistically distinguishable at equal sample sizes, but the mean relative errors for these features remained small (7–15%), and the detected differences likely reflected SA's tendency to compress distributional tails rather than shift means. This tail compression arises from two sources: the PCA dimensionality reduction, which cannot fully represent higher-order moments of every feature, and the direction-magnitude decomposition, which draws magnitudes from a finite empirical distribution of K=23 norms. These results represent an honest trade-off: SA sacrifices some tail fidelity in exchange for preserving the manifold geometry that parametric methods cannot capture in the $n < p$ regime.

To assess whether more expressive generative models could match SA at this sample size, we compared against CTGAN and TVAE [Xu et al., 2019], two widely used deep generative methods for tabular data (Supplementary Table S6). CTGAN failed catastrophically, producing median MREs of ≈19%—an order of magnitude worse than SA—across all epoch counts tested (300–3,000), consistent with known GAN mode collapse behavior on small datasets. TVAE achieved comparable marginal fidelity to SA at high epoch counts (median MRE 1.8% at 3,000 epochs), but operated on per-visit rows (n=90) rather than concatenated longitudinal profiles (n=23), meaning it could not capture cross-visit covariance structure and had no mechanism for conditional subgroup generation. These results confirmed that SA's advantage at small n is not merely theoretical: standard deep generative methods either fail outright (CTGAN) or achieve marginal-level fidelity without the longitudinal and conditional capabilities that SA provides.

Several additional limitations warrant discussion. Our study used a single longitudinal dataset of K=23 patients, and further evaluation on independent datasets and across different clinical domains is needed to establish generalizability. The PCA dimensionality reduction assumed linear structure in the feature space; nonlinear manifold learning methods may better capture complex clinical relationships, though at the cost of reduced interpretability. The mechanistic validation under TF+TM conditions revealed systematic ODE model bias, though the observation that real and synthetic patients shared the same bias pattern was itself informative. Finally, we did not validate synthetic patients against clinical outcomes—our validation was restricted to statistical and mechanistic plausibility, and prospective validation against real clinical decision-making remains an important future direction.

5 Conclusion

We have demonstrated that Stochastic Attention can generate synthetic longitudinal patient data from very small clinical cohorts (K=23) that pass four levels of validation: marginal plausibility, cross-visit covariance preservation, conditional subgroup amplification, and mechanistic consistency with an independent ODE model of the coagulation cascade. The SA framework preserves the geometry of the clinical data manifold in regimes where parametric approaches such as multivariate normal sampling fail due to rank deficiency. Combined with a rigorous multi-level validation framework, this approach provides a practical path for generating clinically validated synthetic cohorts to support computational research in rare conditions and small longitudinal studies.

Data Availability

The code for synthetic patient generation, all validation scripts, and the generated synthetic datasets are available at `https://github.com/varnerlab/SA-generation-legacy-dataset-paper`. The real patient data were collected under IRB approval at the University of Vermont Medical Center; de-identified data are available upon reasonable request to the corresponding author.

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Author Contributions

J.D.V. conceived the study, developed the SA pipeline and validation framework, performed computational analyses, and wrote the manuscript. M.B. and T.O. collected the clinical data, provided domain expertise on coagulation biology and thrombin generation assays, and reviewed the manuscript. I.B. designed and oversaw the clinical study, provided clinical interpretation, and reviewed the manuscript. All authors approved the final version.

References

Sarah Toy. U.S. maternal mortality hits highest level since 1965. The Wall Street Journal, 2023. March 16, 2023.

Emily Harris. US maternal mortality continues to worsen. *JAMA*, 329(15):1248, 2023. doi: 10.1001/jama.2023.5254.

Susanna L Trost, Jennifer Beauregard, Fanta Njie, Joyita Berry, Allison Harvey, David A Goodman, Emily E Petersen, and Shanna Cox. Pregnancy-related deaths: Data from maternal mortality review committees in 36 US states, 2017–2019, 2022. `https://www.cdc.gov/maternal-mortality/php/data-research/`.

Kenneth G Mann, Kathleen Brummel-Ziedins, Thomas Orfeo, and Saulius Butenas. Thrombin generation in hemorrhage control and vascular occlusion. *Circulation*, 124(2):225–235, 2011. doi: 10.1161/CIRCULATIONAHA.110.952648.

Saulius Butenas, Thomas Orfeo, and Kenneth G Mann. Tissue factor in coagulation: Which? where? when? *Arteriosclerosis, Thrombosis, and Vascular Biology*, 29(12):1989–1996, 2009. doi: 10.1161/ATVBAHA.108.177402.

Mark F Hockin, Kenneth C Jones, Stephen J Everse, and Kenneth G Mann. A model for the stoichiometric regulation of blood coagulation. *Journal of Biological Chemistry*, 277(21):18322–18333, 2002. doi: 10.1074/jbc.M201173200.

Margareta Hellgren. Hemostasis during normal pregnancy and puerperium. *Seminars in Thrombosis and Hemostasis*, 29(2):125–130, 2003. doi: 10.1055/s-2003-38897.

E Grouzi, A Pouliakis, A Aktypi, et al. Pregnancy and thrombosis risk for women without a history of thrombotic events: a retrospective study of the real risks. *Thrombosis Journal*, 20(60), 2022. doi: 10.1186/s12959-022-00419-6.

Katarina A Bremme. Haemostatic changes in pregnancy. *Best Practice & Research Clinical Haematology*, 16(2):153–168, 2003. doi: 10.1016/S1521-6926(03)00021-5.

Ricardo Azziz, Keslie S Woods, Rosario Reyna, Thomas J Key, Eric S Knochenhauer, and Bulent O Yildiz. The prevalence and features of the polycystic ovary syndrome in an unselected population. *Journal of Clinical Endocrinology & Metabolism*, 89(6):2745–2749, 2004. doi: 10.1210/jc.2003-032046.

Edgardo Abalos, Cristina Cuesta, Ana L Grosso, Doris Chou, and Lale Say. Pre-eclampsia, eclampsia and adverse maternal and perinatal outcomes: a secondary analysis of the World Health Organization multicountry survey on maternal and newborn health. *BJOG: An International Journal of Obstetrics & Gynaecology*, 121:14–24, 2014. doi: 10.1111/1471-0528.12629.

Maria C Bravo, Thomas Orfeo, Kenneth G Mann, and Stephen J Everse. Modeling of human factor Va inactivation by activated protein C. *BMC Systems Biology*, 6:45, 2012. doi: 10.1186/1752-0509-6-45.

Kathleen E Brummel-Ziedins, Thomas Orfeo, Peter W Callas, Matthew Gissel, Kenneth G Mann, and Edwin G Bovill. The prothrombotic phenotypes in familial protein c deficiency are differentiated by computational modeling of thrombin generation. *PLoS ONE*, 7(9):e44378, 2012. doi: 10.1371/journal.pone.0044378.

Deyan Luan, Michael Zai, and Jeffrey D Varner. Computationally derived points of fragility of a human cascade are consistent with current therapeutic strategies. *PLoS Computational Biology*, 3(7):e142, 2007. doi: 10.1371/journal.pcbi.0030142.

Deyan Luan, Fania Szlam, Kenichi A Tanaka, Philip S Barie, and Jeffrey D Varner. Ensembles of uncertain mathematical models can identify network response to therapeutic interventions. *Molecular BioSystems*, 6(11):2272–2286, 2010. doi: 10.1039/b920693k.

Adarsh Gonzales, Guruprabha Guruswamy, and Sandesh R Smith. Synthetic data generation methods in healthcare: A review on open-source tools and methods. *Computational and Structural Biotechnology Journal*, 23:2892–2903, 2023. doi: 10.1016/j.csbj.2024.07.005.

Hasan Murtaza, Nicholas Schreck, Jil Gao, Andreas D Meid, Thomas Gredner, Walter E Haefeli, Hermann Brenner, and Ben Schottker. Synthetic data generation for a longitudinal cohort study – evaluation, method extension and reproduction of published data analysis results. *Scientific Reports*, 14:12785, 2024. doi: 10.1038/s41598-024-62102-2.

Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004. doi: 10.1016/S0047-259X(03)00096-4.

Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Advances in Neural Information Processing Systems*, volume 27, pages 2672–2680, 2014.

Diederik P Kingma and Max Welling. Auto-encoding variational Bayes. In *International Conference on Learning Representations (ICLR)*, 2014.

Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling tabular data using conditional GAN. 32, 2019.

Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Milena Pavlović, Geir Kjetil Sandve, Victor Greiff, David Kreil, Michael Kopp, Gunter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations (ICLR)*, 2021.

Jeffrey D Varner. Multiplicity-weighted stochastic attention for conditional sequence generation, 2024. Available at `https://github.com/varnerlab/SA-Binding-Generation-Study`.

Maria Bravo, Thomas Orfeo, Ira Bernstein, Sandra Vadhin, Divya Lakshmi, and Jeffrey D Varner. BSTModelKit.jl: Building biochemical systems theory models of individualized coagulation dynamics during pregnancy, 2023. JuliaCon 2023.